Quiz 3 - DSC 10, Winter 2026

Full Name:

PID:

Quiz Time: ◯ 1PM ◯ 2PM

Instructions:

- This quiz consists of 5 questions. You have a total of 30 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.
- Assume we have already run `import babypandas as bpd` and `import numpy as np`.

Person to your left:

Person to your right:

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version B

Please do not open your quiz until instructed to do so.

Ray and Minchan have a joint playlist that combines all songs they both listened to in the past month. Each row in the `songs` DataFrame represents a song in the playlist. The columns are: `song_name` (str), `artist` (str), `genre` (str), as well as two columns for the amount of times either has played each song, `plays_ray` (int) and `plays_minchan` (int). Below are the first six rows of songs, though there are many more entries.

	song_name	artist	genre	plays_ray	plays_minchan
0	Put Your Records On	Corinne Bailey Rae	Pop/Soul	78	26
1	On Melancholy Hill	Gorillaz	Alt/Indie	39	121
2	Maps	Yeah Yeah Yeahs	Rock/Alt	23	163
3	Perfect	One Direction	Pop	114	18
4	Swimming Pools (Drank)	Kendrick Lamar	Hip-Hop/Rap	143	45
5	Landslide	Fleetwood Mac	Rock/Folk	92	97

Question 1

Minchan wants to see his listening trends. Suppose he takes many samples of size 30 from `plays_minchan` and computes their means. For each statement below, select the best answer.

a) We expect that at least 75% of all of the sampled values of `plays_minchan` lie within 2 standard deviations of the population mean.

◯ Yes, due to CLT
◯ Yes, due to Chebyshev's
◯ No, due to CLT
◯ No, due to Chebyshev's
◯ Yes, but for reasons that have nothing to do with Chebyshev's or the CLT
◯ No, but for reasons that have nothing to do with Chebyshev's or the CLT

b) We expect that approximately 95% of all sample means lie within 2 standard errors of the population mean.

◯ Yes, due to CLT
◯ Yes, due to Chebyshev's
◯ No, due to CLT
◯ No, due to Chebyshev's
◯ Yes, but for reasons that have nothing to do with Chebyshev's or the CLT
◯ No, but for reasons that have nothing to do with Chebyshev's or the CLT

c) Now assume the distribution of `plays_minchan` is right-skewed but with no outliers. Which of the following statements is true about these statistics of `plays_minchan`?

◯ The median is always greater than the mean
◯ The mean is always greater than the median
◯ The mean is usually greater than the median, but not always
◯ The median is usually greater than the mean, but not always

d) If both Minchan's and Ray's plays distributions are normal, will the distribution formed by pooling both of their plays into one combined dataset be approximately normal?

◯ Always ◯ Sometimes ◯ Never

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Question 2

Select all sampling methods below where the resulting sample is an SRS of 20 songs from the `songs` DataFrame.

☐ Use `songs.sample(20, replace = 1)`

☐ Use `np.random.multinomial(20, np.ones(songs.shape[0]) / songs.shape[0])` to decide how many times to take each row for a sample.

☐ `np.random.choice(songs.get("song_name"), 20)`

☐ Randomly pick 20 genres without replacement, then within each picked genre, randomly pick a song.

☐ Randomly pick 20 artists without replacement, then within each picked artist, randomly pick a song.

☐ None of the above.

Question 3

a) Suppose Ray collected an SRS of the whole playlist (of an unspecified size), storing it in DataFrame `ray_playlist`. Complete the code below to bootstrap for an 80% confidence interval for the mean:

```
boot_means = np.array([])

for i in range(1000):
    sim = ray_playlist.___(a)___.get("plays_ray").mean()
    boot_means = np.append(boot_means, sim)

left = np.percentile(boot_means, ___(b)___)
right = np.percentile(boot_means, ___(c)___)
```

(a): _____

(b): _____

(c): _____

b) After generating 1000 bootstrap means from `ray_play`, what does the distribution of `boot_means` approximate?

◯ The sampling distribution of the sample mean of plays by Ray
◯ The population distribution of plays by Ray
◯ The empirical distribution of plays by Ray
◯ The probability distribution of plays by Ray

c) Which statements are correct about the 80% bootstrap confidence interval of the population mean that Ray just computed?

☐ 80% of songs have a number of plays by ray that lies within the interval.
☐ There is a 80% probability that the true mean lies in this interval.
☐ 80% of such intervals created from repeating the bootstrap process with new samples will contain the true mean.
☐ 80% of the bootstrap means lie in the interval.

Question 4

a) Ray calculated a 95% CLT-based confidence interval for the population mean of his number of plays per song, spanning [60.5, 100.5]. If the sample size used to construct the interval was 25 songs, what is the standard deviation of Ray's plays?

[ ] ◯ Not enough information.

b) If Ray increases the sample size from 25 songs to 100 songs, how does the width of his 95% CLT-based confidence interval change?

◯ Doubles ◯ Halves ◯ Becomes 1/4 as wide ◯ Stays the same

c) Now Ray is curious about other confidence intervals he could have constructed. Given `stats.norm.cdf(1.75) = 0.96`, calculate the endpoints of a 92% CLT-based confidence interval.

[ ] ◯ Not enough information.

d) Ray then constructs a 99% CLT-based confidence interval. Select all the cases that could be possible when comparing his new 99% CI to his original 95% CI.

☐ 99% CI is wider than 95%
☐ 99% CI is smaller than 95%
☐ They have the same width

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Reminders:

- **Write your PID** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.